DOC-20 | CONTROLLED SOURCE SYNCHRONIZATION

Cross-reference-only revision notice

v1.0-RC2 | 11 September 2026 | SOURCE-OF-TRUTH SYNCHRONIZATION ONLY

Control	Selected value
Revision	v1.0-RC1 -> **v1.0-RC2**
Semantic change	**NONE.** This revision changes source version/count/path references only.
Wave-2 semantic authorities	DOC-07 RC2 | DOC-08 RC2 | DOC-15 RC8 | DOC-16 RC5
Canonical HTTP companion	contracts/http/BLOG-DOC-15_openapi_v1.0-RC8.yaml
OpenAPI SHA-256	bf577a463b4c5d8fdceddc2087f53216426ef88d67d8e1a0d22ccf819751cd00
RC8 estate	113 operations | 86 paths | 221 schemas/helpers | 1,311 resolved internal refs | 0 unresolved refs 12 AUTH | 2 PUBLIC/PREVIEW | 63 SITE | 36 GLOBAL

DOCUMENT-SPECIFIC SYNCHRONIZATION

• Operational/performance references now consume DOC-15 v1.0-RC8 and DOC-16 v1.0-RC5.

• Where the inherited body asserts the HTTP estate/count gate, substitute the accepted RC8 estate: 113 operations | 86 paths | 221 schemas/helpers | 1,311 resolved internal refs | 0 unresolved refs; 12 AUTH | 2 PUBLIC/PREVIEW | 63 SITE | 36 GLOBAL.

• No latency, availability, rate-limit, capacity, backup, RPO/RTO, DR or observability target is changed. Health probes remain outside the canonical /api/v1 product operation estate.

SUPERSESSION AND HISTORICAL CLASSIFICATION

The inherited body that follows is retained unchanged in substantive page content. Any inherited non-owning reference to DOC-15 RC7/RC6, the old 112/85/217/1,297 estate, 11 AUTH operations, DOC-16 RC4, or the RC7 YAML filename is classified **HISTORICAL / SUPERSEDED REFERENCE** and is not normative in this revision. All other inherited owned semantics remain in force.

Occurrence ledger: RC7/old HTTP: inherited printed pages 1, 3, 21, 22; RC4: inherited printed pages 1, 3, 21.

IMPLEMENTATION IMPACT: NONE. The accepted NAKSHATRA_MASTER_WAVE2_HANDOFF application/database/API behavior is unchanged by this document synchronization.

BLOG-DOC-20

Performance, Scale, Concurrency, Backup, Disaster Recovery & Observability

Developer execution authority for measurable load behavior, horizontal scale, capacity protection, operational telemetry, backups and disaster recovery across the centralized 14-site blogging platform.

14	1	1	1
isolated public sites	shared API/backend core	PostgreSQL authority	site-aware worker/cache estate

CORE OPERATIONAL PRINCIPLE

Scale shared infrastructure horizontally without weakening tenant isolation, authorization freshness, RLS, deterministic concurrency or derived-state correctness. PostgreSQL remains canonical. Redis, queues, search, sitemap and derivative media remain rebuildable/operational state according to their owning documents.

Document control	Value
Status	REVIEW CANDIDATE - built against user-designated finalized/frozen BLOG-DOC-01 through BLOG-DOC-19; freeze after DOC-20 approval
Version	1.0-RC1
Date	10 September 2026
Upstream authorities	BLOG-DOC-01 through BLOG-DOC-19, including DOC-15 RC7, DOC-16 RC4, DOC-17 RC2, DOC-18 RC2 and DOC-19 RC2
Primary authority	Performance targets; load/scale behavior; stateless replica rules; DB pool/capacity budgets; noisy-neighbor controls; load testing; observability thresholds; backup/PITR policy; RPO/RTO; restore drills
Explicit non-authority	Product features/permissions/pages (01-06); content lifecycle (07); schema/RLS/index definitions (08); routing (09); page flows/data/frontend/API (10-15); SEO/search algorithms (16); media transform/CDN semantics (17); queue/job/cache mechanics (18); credential/security/audit semantics (19); concrete stack/deployment topology (23-24)

DEVELOPER LOCK TARGET

A competent developer must not guess latency objectives, how many connections replicas may consume, which dependency failures make a process unready, whether PITR is used, what is backed up versus rebuilt, how tenant fairness is protected under load, which metrics/alerts are release-critical, or how often restore drills must prove RPO/RTO.

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1. Purpose, Authority and Conflict Rule

BLOG-DOC-20 is the operational performance and recoverability authority for the platform defined by DOC-01 through DOC-19. It converts the already-frozen architecture into measurable service objectives, resource-usage constraints, observability rules and disaster-recovery evidence without changing business behavior.

CONFLICT RULE

If a performance optimization requires changing an upstream permission, PageData field, endpoint, Post state, storage ownership rule, queue/job contract, Redis key semantic, security requirement or database schema decision, optimization stops and the owning upstream document is reopened. Performance never wins by weakening correctness or isolation.

1.1 One source of truth per decision

Decision	Authority
Product/site/application topology	DOC-01 / DOC-14
Authorization/SiteContext/RLS	DOC-02 / DOC-04 / DOC-08
Content lifecycle/concurrency semantics	DOC-07
Tables/constraints/baseline indexes	DOC-08
HTTP contract/rate ceilings	DOC-15 RC7
SEO/search/sitemap computation	DOC-16 RC4
Media transform/public CDN semantics	DOC-17 RC2
Queues/jobs/retries/Redis keys and TTLs	DOC-18 RC2
Security/session/audit/privacy semantics	DOC-19 RC2
Performance/SLO/backup/DR/observability thresholds	DOC-20 - this document
Executable cross-system QA	DOC-21 later
Concrete technology/repository/runtime topology	DOC-23 later
Deployment/environment/rollback mechanics	DOC-24 later

2. Frozen Upstream Contract and Inherited Invariants

Authority	Inherited operational lock
DOC-01	One shared platform with 14 independent public apps; shared infra never means shared ownership.
DOC-03	100,000+ Post-scale direction and public effects after publish/update/archive/tenant lifecycle.
DOC-04	Every site operation resolves trusted SiteContext; pooled DB connection cannot carry stale tenant authority; cache/search/media/worker isolation is release-blocking.
DOC-07	Optimistic Post versioning and publish/schedule invariants are correctness fences, not tunable away.
DOC-08	33 canonical tables, 20 SITE RLS tables, required indexes and transaction-local SET LOCAL app.site_id. No new performance table is created here.
DOC-11/13	Public responses preserve PageData semantics; performance caching cannot add stale/wrong-site fields.
DOC-15 RC7	112 operations and exact transport ceilings/statuses remain unchanged; health probes are operational, not new /api/v1 business endpoints.
DOC-16 RC4	Search ranking/indexability/sitemap algorithms are frozen; DOC-20 sizes

Authority	Inherited operational lock
	them only.
DOC-17 RC2	Private original, sanitized native_webp baseline, fixed derivative catalog and CDN revocation semantics remain frozen.
DOC-18 RC2	Six logical queues, 11 jobs, retries, dedupe state machine, leases, exact Redis namespace/TTLs and reconciliation remain frozen.
DOC-19 RC2	Security/session/CSRF/CORS/secrets/audit/privacy semantics and retention minima remain frozen; DOC-20 supplies operational thresholds and backup/DR budgets.

DERIVED-STATE DOCTRINE

A cache hit, search document, sitemap artifact, media derivative, queue record or observability record can improve speed/operations but cannot become editorial truth. Recovery reconstructs derived state from the correct authoritative source instead of writing stale derived state back into PostgreSQL.

3. Operational Architecture and Scaling Doctrine

```
14 independent public frontends -> CDN/edge -> shared stateless API replicas
                                  |
                                  +-> PostgreSQL primary (canonical)
                                  +-> Redis (cache/coordination)
                                  +-> Search adapter (derived)
                                  +-> Object storage/CDN (DOC-17)
                                  +-> Durable broker -> worker pools (DOC-18)

Observability plane: logs + metrics + traces + alerts
Recovery plane: encrypted backups + WAL/PITR + object protection + restore drills
```

- API and worker processes scale horizontally; correctness must remain valid with one replica or many.
- No replica may hold authoritative tenant/session/permission/Post state only in local memory between requests.
- PostgreSQL primary remains the transactional source for canonical tables. Optional replicas are acceleration only and are constrained in Section 8.4.
- Redis loss may reduce performance or force rebuild/replay but never authorizes a request or rewrites canonical truth.
- Capacity is shared but resource admission must prevent one hot site from starving the remaining sites when other tenants have queued/active work.
- Deployment provider, exact instance types and orchestrator are deliberately left to DOC-23/24; this document freezes measurable behavior they must satisfy.

4. Measurement Model, SLO Vocabulary and Availability Accounting

4.1 Measurement rules

Term	Locked meaning
p50/p95/p99	Percentile over successful eligible requests in the measurement window; retries are separate requests and not hidden.
Origin service time	Server receive through first response byte from application/gateway, excluding end-user network latency.
Edge time	Synthetic client-to-CDN response timing from approved test vantage; used only for CDN checks, not API-origin SLO calculation.
Availability	Eligible request would be served by the platform and does not fail because of a platform-controlled dependency.

Term	Locked meaning
Excluded from outage denominator	Intentional 301/302/308/404/410, 401/403, 409, 422, 429 caused by contract enforcement, and client disconnects before request completion.
Counted as unavailable	Unexpected 5xx, dependency 503 for a normally healthy surface, readiness-removal that makes required capacity unavailable, or timeout beyond the defined service budget.
Healthy-system derived SLO	Measured when required dependencies are available; outage recovery is separately measured by RTO/reconciliation.

4.2 Latency target registry

ID	Surface	p95	p99
SLO20-LAT-01	Public PageData - warm Redis/origin cache	p95 <= 200 ms	p99 <= 500 ms
SLO20-LAT-02	Public PageData - uncached origin	p95 <= 600 ms	p99 <= 1,200 ms
SLO20-LAT-03	Public search request	p95 <= 800 ms	p99 <= 1,500 ms
SLO20-LAT-04	Admin read/list/search	p95 <= 700 ms	p99 <= 1,500 ms
SLO20-LAT-05	Admin non-media mutation	p95 <= 1,000 ms	p99 <= 2,000 ms
SLO20-LAT-06	Auth login/MFA verification round trip	p95 <= 1,500 ms	p99 <= 3,000 ms
SLO20-LAT-07	Async command acceptance, including reindex/audit export	p95 <= 500 ms	p99 <= 1,000 ms
SLO20-LAT-08	Media upload/replace synchronous acceptance phase	p95 <= 10 s	p99 <= 20 s under reference image workload

LATENCY INTERPRETATION

These are origin/application budgets under the reference profile in Section 5. They do not redefine browser LCP or site-specific visual performance, which depends on each independent frontend. A request that is rate-limited or denied correctly is not made slower merely to satisfy a percentile.

5. Reference Capacity and Load-Test Profile

DOC-20 freezes a reproducible verification profile, not a lifetime software ceiling. Production sizing may exceed it without changing contracts.

Dimension	Reference profile
Sites	All 14 initial sites ACTIVE; same-slug and same-filename collisions intentionally present across tenants.
Canonical content	At least 100,000 Posts total, with realistic Category/Tag/Author/Media relationships and published/archive/trash mix.
Hot-tenant distribution	SITE-A receives 60% of public traffic, SITE-B 20%, remaining 12 sites share 20%; correctness must remain isolated.
Public steady load	300 requests/second for 30 minutes.
Public burst	600 requests/second for 5 minutes without correctness loss; scaling may increase replicas.
Public search	20 requests/second mixed queries within the public load.
Admin steady/burst	25 req/s steady; 50 req/s for 5 minutes across reads and permitted mutations.

Dimension	Reference profile
Media	5 simultaneous uploads/replacements using valid images up to 20 MiB; variant processing continues asynchronously per DOC-17/18.
Scheduled publish spike	140 Posts due within the same 10-second window, distributed across all sites.
Worker backlog	10,000 mixed non-destructive derived jobs across sites, including hot-tenant skew.
Failure injectors	One API replica loss, Redis restart, worker restart, search adapter unavailable, broker latency, object-store transient failure and DB failover simulation.

6. Stateless API Replica Model and Horizontal Scaling

6.1 Statelessness contract

- No request may depend on in-process memory from a prior request for tenant authority, permission, session validity, idempotency outcome, Post version, Preview authority, cache epoch or worker acceptance.
- Local process memory may contain immutable configuration, compiled schema, bounded connection-pool state and ephemeral non-authoritative caches only. A local cache miss/loss must never change correctness.
- Session/step-up/Preview security state follows DOC-19 logical stores and is not pinned to one API replica.
- Any replica receiving the same valid request can reconstruct the same SiteContext and authorization result from shared authoritative/security state.

6.2 Horizontal-scaling trigger policy

Signal	Scale-up trigger	Scale-down guard
CPU	>70% average for 5 min	<45% for 15 min
Request concurrency	>70% of configured per-replica safe concurrency for 3 min	<40% for 15 min
p95 origin latency	Above applicable SLO for 5 min with dependency health normal	Below 80% of SLO for 15 min
Pool wait	p95 >100 ms for 5 min and DB not saturated	p95 <25 ms for 15 min

SCALE SAFETY

Autoscaling may add replicas only if aggregate database/broker/Redis connection budgets remain inside the limits in Sections 8, 11 and 13. Scaling the API must not create a database connection storm.

7. Noisy-Neighbor Protection and Admission/Fairness

Transport rate ceilings remain DOC-15/19 authority. DOC-20 adds internal capacity fairness so a hot tenant cannot consume shared pools indefinitely while another tenant has eligible work.

Layer	Fairness rule
API admission	When shared capacity is saturated, preserve per-site admission slots for active tenants rather than allowing one site to monopolize every execution slot. Do not raise or bypass DOC-15 ceilings.
PostgreSQL pool	Background work uses a separately budgeted pool and may not exhaust all connections needed by interactive API traffic.
Workers	If more than one site has queued work, one site must not consume >50% of a shared queue worker pool for >60 seconds, except where DOC-18 serialization leaves fewer runnable jobs.

Layer	Fairness rule
Search reindex	One full reindex per site; global worker capacity must preserve incremental work for other sites.
Media	Large media backlog for one site must not block scheduled publishing or cache invalidation because logical queues remain separate.
Redis	Keys are already tenant-namespaced; memory policy must not use one tenant as authorization or correctness boundary.
CDN	Origin protection/rate shaping may defend hot objects but never mix cache identity between sites.

NO DEGRADATION SHORTCUT

Noisy-neighbor protection may queue, shed, or return existing 429/503 outcomes where the owning HTTP/security contract allows. It may not return another site's cached data, skip authorization, disable RLS, publish early, or serve private media.

8. PostgreSQL Topology, Pools and Connection Budgets

8.1 Primary authority

The production PostgreSQL primary is the canonical writer for DOC-08 tables. All mutations and authorization-sensitive reads that require current truth execute against the primary unless a later controlled revision explicitly proves equivalent correctness.

8.2 Aggregate connection budget

Budget	Rule
Total application budget	Sum of maximum API + worker + operational pools MUST be <= 70% of database max_connections.
Reserved headroom	At least 30% remains for failover, migrations/maintenance, observability, emergency access and burst protection.
API replica pool	Maximum 20 connections per replica; minimum pool may be 2. Autoscaling controller must respect aggregate 70% ceiling before adding replicas.
Worker replica pool	Maximum 10 DB connections per worker process/replica; long-running batch workers use bounded batch concurrency.
Pool checkout timeout	3 seconds for interactive API; 10 seconds for workers before controlled retry/failure.
Idle connections	Pools must retire idle connections according to driver/provider guidance and must not retain tenant context.
Tenant setup	Every site-owned transaction executes SET LOCAL app.site_id from current trusted SiteContext exactly as DOC-08 requires.

8.3 Pool saturation behavior

- Pool exhaustion must queue briefly within checkout timeout; it must not open unbounded ad hoc connections.
- Interactive Admin/Public requests take precedence over non-due maintenance scans when connection pressure is high.
- Workers that fail checkout treat it as TRANSIENT under their existing DOC-18 retry profile; DOC-20 does not change attempt counts/backoff.
- Returning a connection to the pool after a site transaction must leave no persistent app.site_id/session-local tenant state.

8.4 Optional read replicas

NOT A V1 CORRECTNESS REQUIREMENT

Read replicas are OPTIONAL performance accelerators. If adopted, they may serve explicitly stale-tolerant public discovery

reads only after same-site correctness testing. They MUST NOT serve mutations, permission/assignment checks, login/session validation, Preview, Post edit reload before mutation, expected_version checks, publish/schedule decisions, or any read whose staleness could change authorization or lifecycle behavior.

9. PostgreSQL Query/Index Performance and Statement Budgets

Class	Budget / rule
Interactive simple/indexed read	p95 DB execution <= 150 ms; p99 <= 500 ms under reference dataset.
Interactive write transaction	p95 DB execution <= 300 ms excluding Argon2/media processing; p99 <= 800 ms.
Interactive SQL statement timeout	5 seconds read; 10 seconds write. Exceeding returns/propagates controlled failure, never partial commit.
Background batch statement timeout	30 seconds per statement; jobs chunk work so a 30-minute job does not run one 30-minute SQL statement.
Baseline indexes	All DOC-08 required indexes remain present. Performance work may not drop/replace them silently.
Additional index	Only through measured EXPLAIN/ANALYZE evidence and controlled DOC-08/DOC-24 migration review; no runtime CREATE INDEX.
Query shape	Site-owned high-volume queries include site_id in predicate/index path and never fetch all sites then filter in application.

10. Database Concurrency, Locks and Transactional Correctness

- DOC-07 expected_version and DOC-08 compare-and-swap remain the final lost-update guard for versioned Post mutations; higher throughput cannot replace them with last-write-wins.
- Deadlock/serialization failures are safe retry candidates only when the owning API idempotency and worker idempotency contract permits retry. A retry must reauthorize/revalidate current state.
- Interactive transactions should hold locks for the shortest practical duration; external network calls, object processing and queue waiting must not occur while holding canonical content-row locks unless the upstream transaction explicitly requires it.
- Long maintenance scans are chunked by stable keys and yield between batches to protect interactive latency.
- Database failover or connection reset must abort in-flight transactions. The application never assumes an unknown transaction committed; idempotency/authoritative reload resolves ambiguity.

Concurrency signal	Threshold
DB lock wait p95	Warning >100 ms for 10 min; critical >500 ms for 5 min
Deadlock rate	Critical if >5/min for 5 min or any sustained tenant-correlated spike
Pool utilization	Warning >80% for 10 min; critical >95% for 5 min
Connection count	Critical at >90% of max_connections; autoscaling API blocked before this point

11. Redis Capacity, HA and Cache Performance

DOC-18 owns exact Redis keys and TTLs. DOC-20 owns availability/capacity behavior around that frozen key contract.

Concern	Locked operational rule
Availability	Production Redis uses HA/failover capable topology. Exact product/provider belongs DOC-23/24.

Concern	Locked operational rule
Memory headroom	Steady-state used memory <=70% of configured usable memory; warning at 75%, critical at 90%.
Eviction	No correctness-critical coordination key may rely on best-effort eviction. Any unexpected eviction event is alertable; coordination keys with no TTL are explicitly protected by capacity sizing.
Latency	Command p95 <=5 ms and p99 <=20 ms from application network under normal load.
Cache hit objective	Repeated eligible public-route synthetic test reaches >=90% cache hit after warm-up; organic hit rate is observed, not a product correctness gate.
Failure mode	Redis outage disables/loses cache/coordination acceleration but must not broaden access. API falls back to authoritative reads where safe; worker dedupe loss may create duplicates but not missed canonical effects per DOC-18.
Persistence	Redis persistence/replication may reduce operational recovery time but is not canonical backup. Search status/sitemap counters are reconstructable.

12. CDN, Public Delivery and Edge Performance

Area	Rule
Public page/static delivery	CDN/reverse proxy may cache only responses allowed by DOC-15/18/19; Admin and Preview success remain no-store.
Media CDN	DOC-17 remains authority for sanitized public baseline, variant URLs, private original protection, immutable/revisioned locators and purge/revocation.
Cache identity	Host/site remains part of cache identity. Same path on two sites can never share a tenant-blind response cache object.
Origin shielding	Enable provider-supported origin shielding/coalescing when available so hot objects do not stampede API/object origin.
Compression	Textual public assets/responses should use negotiated compression where safe; representation semantics remain unchanged.
Edge failure	If CDN is unavailable, fail over to protected origin path only if the deployment contract preserves canonical host, TLS, security headers and tenant resolution.
Stale content	No stale-if-error rule may bypass site disable/archive, Preview/private eligibility, media revocation or redirect/canonical security decisions.

12.1 Edge synthetic targets

Target	Threshold
Cached public static/media asset	p95 <=250 ms from approved synthetic vantage
CDN origin error ratio	<1% over 5 min under normal dependency health
Origin request amplification	<=1.2 origin requests per unique cache miss burst after coalescing/warm-up

13. Worker and Queue Scaling

DOC-18 retry, dedupe, timeout, lease and serialization mechanics are immutable. DOC-20 scales capacity around them.

Queue family	Scale signal	Operational capacity objective
q18.publish	due-job age + queue depth	Under reference spike, >=99% eligible due

Queue family	Scale signal	Operational capacity objective
		publishes transition within 60s of scheduled time.
q18.media	oldest age + CPU + object-transform throughput	No starvation of publish/cache; remaining variants p99 converge within 15 min under reference load.
q18.search	oldest incremental age + depth	Incremental search p99 <=120s; reserve capacity during full reindex.
q18.sitemap	desired_seq-built_seq lag	One active build/site as DOC-18; p99 site rebuild <=15 min.
q18.cache	oldest age + depth	Cache invalidation p99 <=30s; this queue receives priority over maintenance.
q18.maintenance	reconcile lateness	Low priority; may pause under pressure but must resume before its defined cadence misses two consecutive cycles.

13.1 Worker autoscaling rules

- Scale on oldest-job age first, queue depth second; depth alone is not sufficient when jobs have different execution cost.
- Scale-up cannot exceed downstream DB/object/search/Redis safe concurrency budgets.
- When multiple sites have runnable work, scheduling honors the 50% noisy-neighbor fairness rule from Section 7.
- Scale-down waits at least 10 minutes after queue-age recovery and never terminates a job without broker-safe re-delivery semantics.

14. Scheduled Publishing Timing Objectives

ID	Condition	Objective
SLO20-PUB-01	Healthy dependencies, job already delayed/ready	p95 publish <=10s after scheduled_for_utc; p99 <=60s.
SLO20-PUB-02	Reference spike of 140 due Posts	>=99% eligible Posts publish within 60s; 100% within 5m unless an upstream validation/state change intentionally prevents publication.
SLO20-PUB-03	Delayed queue loss/restart	60-second DOC-18 schedule reconcile repairs missing due work; recovery evidence shows no early/duplicate publish.

CORRECTNESS BEFORE PUNCTUALITY

Missing the timing SLO is an operational incident; it is never corrected by publishing before scheduled_for_utc, skipping DOC-07 eligibility, bypassing expected state/version, or processing a wrong-site Post.

15. Search/Reindex Performance and Scale

Concern	Rule / objective
Public query latency	SLO20-LAT-03: p95 <=800 ms, p99 <=1,500 ms under reference load.
Index isolation	Every query/index build remains same-site; no performance optimization may build one tenant-blind result set then filter.
Incremental convergence	p95 <=30s, p99 <=120s from committed public-affecting mutation to searchable current representation under healthy dependencies.
Single-site full reindex	100,000 eligible documents must rebuild/validate/activate in <=30 min in the reference environment; DOC-18 attempt timeout remains the hard job attempt boundary.

Concern	Rule / objective
Network reindex	If every site requires rebuild, workers preserve site isolation/fairness and complete all sites within 4h reference DR window.
Shadow generation	A failed reindex does not destroy the prior valid generation; activation remains atomic per DOC-18.

16. Sitemap Generation Performance

Metric	Objective
Dirty signal to build start	p95 <=60s when queue healthy
Build + publish per site	p95 <=5m; p99 <=15m under reference content size
Coalescing	Multiple desired_seq increments do not create unbounded concurrent builds; one active build/site
Recovery	After sitemap artifact loss, all ACTIVE-site resources regenerate within 2h platform-wide
Correctness	25,000 URL shard limit/order/eligibility remain DOC-16; performance cannot drop URLs to finish faster

17. Media Processing/CDN Performance

Stage	Objective / invariant
Upload/replace synchronous baseline	SLO20-LAT-08; accepted request has verified private original + sanitized native_webp as DOC-17 requires.
Remaining variants	p95 <=5m; p99 <=15m under reference workload.
Replacement switch	Generation switch remains atomic; old private/public generation is never mixed with new metadata.
CDN purge/revocation	p95 <=30s after authoritative revoke/replace signal; p99 <=2m.
Variant backlog protection	Media workers may scale independently and may not consume publish/cache queue capacity.
Original protection	Backup/recovery never turns canonical private original into anonymous public fallback.

18. Derived-State Convergence SLOs

Derived effect	p95	p99	Authoritative fallback
Public cache invalidation	<=5s	<=30s	Short frozen TTL + PostgreSQL current truth
Incremental public search	<=30s	<=120s	Reconciliation/full reindex
Site sitemap rebuild	<=5m	<=15m	desired_seq/built_seq + reconcile
Remaining media variants	<=5m	<=15m	private original/current generation + reconcile
Scheduled publish	<=10s after due	<=60s after due	Post schedule row + 60s reconcile

19. Health Checks, Readiness and Dependency Degradation

OPERATIONAL SURFACE ONLY

Health probes are infrastructure/orchestrator interfaces, not new DOC-15 business API operations and not part of /api/v1. Exact probe URL/socket command is frozen later by DOC-23/24.

Process	Liveness	Readiness requires	Does NOT require
API	Process/event loop responsive	Configuration/secrets loaded; PostgreSQL primary reachable; pool checkout succeeds; route/site registry query succeeds	Redis/search/object store for unrelated endpoints; those may degrade locally
Auth-capable API	Same	API readiness + logical security store/crypto dependencies needed for auth routes	Search/sitemap/media variant workers
Publish worker	Worker loop responsive	Broker + PostgreSQL + ability to enter CTX-WORKER-SITE	Search/CDN
Media worker	Worker loop responsive	Broker + PostgreSQL + private object store + image processor	Public search
Search worker	Worker loop responsive	Broker + PostgreSQL + search adapter	Media CDN
Sitemap worker	Worker loop responsive	Broker + PostgreSQL + sitemap artifact destination	Media processor
Cache worker	Worker loop responsive	Broker + Redis; CDN purge adapter only for MEDIA_CDN jobs	Search adapter

19.1 Dependency failure rules

- A noncritical dependency failure must not mark every API replica unready if unaffected endpoints can still serve safely.
- A critical dependency failure must remove a replica from readiness before it returns corrupted/tenant-unsafe data.
- Readiness flapping uses a 3-consecutive-failure threshold and 2-consecutive-success recovery unless the dependency reports hard failure.
- Health output exposes only safe component state to the orchestrator; no secrets, site data, connection strings or raw exception details.

20. Structured Logs and Retention

Log class	Retention	Mandatory fields / constraints
Application operational	30 days online	timestamp, service, environment, severity, request_id/correlation_id, safe route/operation ID, outcome, duration; no body/secrets
Worker operational	30 days online	DOC-18 job event semantics, queue/job_type/outcome/duration; no payload secret/content
Security telemetry with raw IP	<=30 days by default	Exactly DOC-19 restriction; restricted access; incident/legal hold required for longer
Secrets access logs	>=90 days	DOC-19 minimum; no secret values
Audit events	>=365 days online	Physical append-only DOC-08/DOC-19 authority; backup retention may temporarily extend inaccessible copies
Restore/DR evidence	365 days minimum	restore point, operator, environment, checksums, measured RPO/RTO, result; no production data dump

20.1 Log safety

- Never log passwords, hashes, MFA secrets, recovery codes, session/Preview bearer values, raw Idempotency-Key, signed private URLs, object-store credentials, ArticleBody, uploaded bytes or full DB connection strings.

- Cross-site denial logs may identify the current safe context/principal internally but do not record guessed target content metadata.
- Sampling may reduce DEBUG/INFO volume but must not sample away ERROR, audit-write failure, backup failure, restore failure or security-critical events.

21. Metrics Registry and Cardinality Rules

ID	Metric family	Allowed dimensions
MET20-API-REQ	api_request_total	operation_id, status_class, scope
MET20-API-LAT	api_request_duration_seconds	operation_id, scope
MET20-API-CONC	api_inflight_requests	scope
MET20-DB-POOL	db_pool_in_use/db_pool_wait_seconds	pool_class
MET20-DB-QRY	db_query_duration_seconds	query_family
MET20-DB-LOCK	db_lock_wait_seconds/deadlock_total	query_family
MET20-REDIS	redis_command_duration_seconds	command_family
MET20-REDIS-MEM	redis_memory_used_ratio	cluster
MET20-CDN	cdn_origin_request_total/cdn_cache_hit_ratio	site_key only where safe internally
MET20-WORKER	worker_job_total/duration/retry_total	queue, job_type, outcome
MET20-QUEUE	worker_queue_depth/oldest_job_age_seconds	queue
MET20-SEARCH	search_query_duration/search_reindex_total	site class/outcome
MET20-SMAP	sitemap_build_total/duration	outcome
MET20-MEDIA	media_variant_job_total/duration	outcome
MET20-BACKUP	backup_last_success_age_seconds/ backup_result_total	asset_class
MET20-PITR	pitr_archive_lag_seconds	database
MET20-RESTORE	restore_drill_duration_seconds/restore_drill_total	scope, outcome
MET20-SEC	audit_write_failure_total/cross_site_denial_total	safe reason/surface

CARDINALITY RULE

Do not use post_id, media_id, full URL, raw query text, email, IP address, session ID, token, Redis key or arbitrary exception text as a metric label. High-cardinality identities belong restricted logs/traces when permitted, not metric labels.

21.1 Metrics retention

Resolution	Retention
1-minute operational series	30 days
5-minute rollup	180 days
1-hour rollup	13 months

Metrics retention is for aggregate operational telemetry only. Security raw-IP data, arbitrary identifiers and request bodies are not converted into metric dimensions to evade DOC-19 privacy limits.

22. Distributed Tracing and Correlation

- X-Request-ID from DOC-15 remains the user-visible correlation identity. Internal trace_id/span_id may be added to telemetry only and are not new API contract fields.

- Propagate trace/correlation across API -> DB -> broker enqueue -> worker -> search/object/CDN adapter where supported; asynchronous causation uses DOC-18 correlation_id/causation_id semantics.
- Sampling: 10% of successful public requests, 25% of successful Admin requests, and 100% of 5xx, timeouts, backup/restore operations, destructive security-sensitive operations and worker DLQ events. Traces never record request/response bodies by default.
- Full-fidelity traces are retained 7 days; aggregate metrics survive longer. Security raw-IP restrictions from DOC-19 still apply.
- If a tracing backend is unavailable, request handling continues safely; telemetry failure must not cause tenant data to be logged elsewhere as compensation.

23. Alerting Thresholds and Severity

ID	Severity	Trigger
ALT20-01	Critical	Unexpected 5xx >5% for 5m on public/admin/auth
ALT20-02	Warning	Unexpected 5xx >2% for 5m
ALT20-03	Warning	Applicable p95 latency exceeds SLO for 10m
ALT20-04	Critical	Applicable p99 latency exceeds 2x SLO for 5m
ALT20-05	Warning	DB pool >80% for 10m or pool-wait p95 >100ms
ALT20-06	Critical	DB pool >95% for 5m or total DB connections >90% max
ALT20-07	Critical	PITR/WAL archive lag >300s or backup job fails
ALT20-08	Critical	No successful base backup in 26h
ALT20-09	Warning	Redis memory >75% for 10m
ALT20-10	Critical	Redis memory >90% or unexpected eviction observed
ALT20-11	Critical	q18.publish oldest eligible due job >60s
ALT20-12	Warning	q18.search or q18.media oldest job >5m
ALT20-13	Critical	q18.search or q18.media oldest job >15m
ALT20-14	Critical	Any new publish DLQ entry or cache invalidation DLQ entry
ALT20-15	Warning	Any other DLQ depth >0 for 10m
ALT20-16	Critical	audit_write_failure_total increases above 0
ALT20-17	Critical	Restore drill fails checksum/integrity/isolation verification
ALT20-18	Critical	Cross-site denial >=5 from same principal/source in 5m or >=50 platform-wide in 5m
ALT20-19	Warning	CDN origin error >1% for 5m
ALT20-20	Critical	Health-ready capacity <50% of required replicas for 5m

24. Availability SLOs and Error-Budget Policy

ID	Surface	Monthly SLO
SLO20-AV-01	Public PageData/API required routes	>=99.90%
SLO20-AV-02	Central Admin protected API	>=99.90%
SLO20-AV-03	Authentication/access HTTP family	>=99.90%
SLO20-AV-04	Queue acceptance for operations whose success	>=99.90%

ID	Surface	Monthly SLO
	requires durable acceptance	
SLO20-AV-05	Public search service	>=99.50% (may degrade without corrupting other public content)

- Error budget is the difference between 100% and the SLO for the calendar month.
- If a surface consumes >50% of its monthly error budget before half the month has elapsed, freeze nonessential performance-affecting rollout for that surface until cause is understood.
- If >100% is consumed, remediation capacity and reliability fixes take precedence over optimization work that increases risk.
- Correctly enforced security/authorization/rate-limit outcomes do not consume availability budget.

25. Backup Architecture and Protection Classes

Class	Canonical?	Protection
PostgreSQL canonical data	YES	Encrypted base backups + continuous WAL/PITR; 35-day rolling retention.
Private media originals	YES for media recovery	Provider durability + replication/backup target; backup copies private; 35-day rolling recovery window.
Media variants	Derived	No mandatory backup; regenerate from current private original/generation.
Redis cache/coordination	Derived	No correctness backup requirement; optional persistence for faster restart only.
Public search index	Derived	No canonical backup requirement; rebuild from PostgreSQL.
Sitemap artifacts	Derived	No canonical backup requirement; rebuild.
Queue messages	Operational durable	Broker durable/HA acceptance; not a substitute for DB backup. Reconciliation repairs inferable derived work.
Frontend build artifacts/config	Deployable	Reproducible from repository/release artifacts under DOC-23/24; not defined as DB backup.
Secrets	Security authority	Secrets-manager version/recovery controls per DOC-19; never copied into ordinary DB/object backups.

26. PostgreSQL PITR Policy

PITR ADOPTION DECISION

Production PostgreSQL PITR IS ADOPTED for v1. Continuous transaction-log/WAL archiving is mandatory. This is a DOC-20 operational decision and does not change DOC-08 schema or application behavior.

Control	Requirement
Base backup frequency	At least every 24 hours.
WAL/archive objective	Archive/replicate continuously; measured archive lag <=5 minutes.
Rolling retention	35 days for recoverable database points, encrypted at rest and in transit.
Backup verification	Daily automated integrity/manifest verification; failure alerts immediately.
Restore target granularity	Select an exact safe timestamp/LSN as supported by the chosen engine; do not restore over production in place without isolated verification.

Control	Requirement
Encryption	Independent backup encryption keys managed by secrets/KMS authority; no key embedded in backup files/source control.
Access	Restricted operations/owner recovery path; ordinary Admin has no backup browser/download capability.
Cross-tenant behavior	A platform DB restore is platform-wide physical recovery; after restore, all site isolation/RLS policies are verified before service readiness.

26.1 PostgreSQL recovery objectives

Metric	Objective
RPO	<=5 minutes for committed canonical DB state
RTO	<=60 minutes to restore validated primary service in the defined DR exercise
Restore validation	Schema/migrations, RLS/FORCE RLS, site registry, permission seeds, Post versions, audit append-only rules and representative cross-site isolation tests all PASS before traffic

27. Object-Storage Backup and Media Recovery

Concern	Rule
Private originals	Canonical private originals require provider-native replicated durability or equivalent recovery protection targeting RPO <=15 minutes.
Recovery window	35-day rolling recoverable version/snapshot window for accidental loss/corruption, subject to privacy/legal deletion rules.
Public variants	Derived and regenerable. They may be restored for speed but are never considered canonical recovery evidence.
Deleted/tombstoned media	Backup copy may remain inaccessible until backup expiry, but must never be re-exposed publicly merely because a historical object version exists.
Restore ordering	Restore media DB ownership/current-generation truth first, then private original availability, then regenerate/verify native_webp and remaining current variants.
Cross-site verification	Every recovered key/object is validated against the owning site/media identity; no filename-based or cross-site object substitution.

28. Derived-State and Queue Recovery Policy

Asset	RPO stance	Recovery rule
Redis cache	No canonical RPO	May be discarded. Rebuild lazily/reconcile; site epochs/keys reinitialize without changing truth.
Redis dedupe/leases	No canonical RPO	Loss can cause duplicate enqueue; DOC-18 idempotency/fences keep duplicates safe. Never infer canonical completion from restored stale dedupe state.
Search index	No canonical RPO	Rebuild same-site index from PostgreSQL; prior validated generation may be retained if trustworthy.
Sitemap	No canonical RPO	Regenerate from DOC-16 current eligibility/routing.
Media variants	No canonical RPO	Regenerate only from current private original/generation.

Asset	RPO stance	Recovery rule
Broker accepted messages	Operational RPO <=5m in catastrophic broker recovery; within normal HA acceptance, acknowledged durable messages must survive process/node restart.	
Security session store	Do not restore stale bearer authority after catastrophic loss; revoke/expire and require reauthentication rather than resurrecting sessions.	

NEVER RESTORE STALE AUTHORITY

Redis/session/queue snapshots are not blindly restored as authorization truth. A DR restore must prefer revocation, reauthorization and source reconciliation over resurrecting stale session, dedupe, lease or tenant-context state.

29. RPO / RTO Matrix

ID	Asset	RPO	RTO
DR20-01	PostgreSQL canonical data	<=5m	<=60m
DR20-02	Private media originals	<=15m	<=2h
DR20-03	API/Admin/Auth processes	0 canonical data loss - stateless	<=30m after infrastructure available
DR20-04	Broker accepted work	<=5m catastrophic / 0 for normal HA restart	<=30m
DR20-05	Redis cache/coordination	No canonical RPO	<=15m service restore; lazy rebuild continues
DR20-06	Public search index	Rebuildable	<=2h site service; <=4h all-site rebuild
DR20-07	Sitemap artifacts	Rebuildable	<=2h all ACTIVE sites
DR20-08	Media derived variants	Rebuildable	<=4h backlog recovery after originals available
DR20-09	Admin/session ephemeral state	Intentionally not restored stale	<=60m auth service; users may re-login

30. Restore and Disaster-Recovery Execution

30.1 Canonical restore sequence

1. Declare incident, stop unsafe writes or isolate affected environment; preserve evidence and choose recovery point within policy.
2. Provision clean recovery infrastructure from approved release configuration; do not repair a compromised primary in place when integrity is uncertain.
3. Restore PostgreSQL base + WAL/PITR to the chosen safe point; verify migration/schema version and database roles.
4. Run database integrity and isolation checks before opening API readiness: sites/domains, RLS/FORCE RLS, permission seed count, representative same-ID/wrong-site access, Post version/state and audit append-only constraints.
5. Restore/verify private media originals to current DB-owned generations; keep all recovered originals private.
6. Redeploy stateless API/Admin/auth processes and security dependencies. Revoke sessions if session-state integrity is not provably current.
7. Reinitialize Redis rather than trusting stale tenant cache/lease/dedupe state; restore only operational state specifically proven safe.
8. Recover broker or start clean broker and run DOC-18 site-scoped reconciliation to reconstruct inferable work. Reissue non-inferable accepted maintenance commands only through existing authorized operations.
9. Rebuild search indexes/sitemaps/media variants from authoritative state, preserving site-by-site isolation.

10. Warm critical public caches gradually; enable traffic in stages while monitoring error/latency/isolation alerts.
11. Close incident only after measured RPO/RTO, audit/security checks, site isolation and derived convergence are recorded.

30.2 Disaster scenarios

Scenario	Required response
Single API replica failure	Remove from readiness; other stateless replicas continue; no session/site affinity required.
PostgreSQL primary failure	Failover/restore per DB topology; abort ambiguous in-flight transactions; clients/workers retry only under frozen idempotency rules.
Redis total loss	Continue authoritative paths where safe; rebuild caches; expect duplicate worker enqueue rather than missed truth.
Broker loss	Restore durable broker within RTO; run reconciliation; no worker may infer completion from stale Redis claims.
Search loss	Public search may degrade; rebuild same-site indexes; other public PageData remains available where safe.
Object-storage regional loss	Restore private originals from protected copy; do not expose historical originals publicly; regenerate sanitized variants.
Security compromise	DOC-19 containment/secret/session rules take priority; DR restore cannot revive revoked secrets or sessions.
Bad release/migration	DOC-24 rollback/migration recovery authority; DOC-20 supplies backup point and verification targets only.

31. Restore Drills and Evidence

Drill	Cadence	Pass condition
Database PITR restore	Quarterly	Restore isolated environment to random point within previous 7 days; measured RPO <=5m and RTO <=60m; integrity/isolation suite passes.
Media-original recovery	Quarterly	Recover representative originals from at least 3 sites, including same filename collision; current ownership/private-only/public derivative regeneration passes.
Redis/search/sitemap rebuild	Quarterly	Start with empty Redis and search/sitemap artifacts; restore service solely from canonical sources within matrix RTO.
Broker/worker restart	Quarterly	Kill/restart broker/worker nodes; accepted durable messages/redelivery/reconciliation preserve correctness.
Full platform DR game day	Every 6 months	Recover DB + media + services in clean environment; prove all 14 sites, Admin auth, background convergence and security/isolation gates.

31.1 Evidence record

```
DR-EVIDENCE {
  drill_id, started_at, completed_at, recovery_point,
  measured_rpo_seconds, measured_rto_seconds,
  database_integrity_result, rls_isolation_result,
  media_private_original_result, derived_rebuild_result,
  security_session_revalidation_result, issues[], owner, approval
}
# Operational evidence only; not a new DOC-08 application table.
```

32. Retention Interaction and Backup-Safe Deletion

Data	Active retention authority	DOC-20 backup interaction
audit_events	DOC-19 >=365 days online; append-only during active retention	35-day backups may contain older copies after online purge; copies remain inaccessible except controlled restore/legal process and expire with backup window.
Raw-IP security telemetry	DOC-19 <=30 days default	Do not place raw security telemetry into a backup set retained longer than the permitted policy unless incident/legal hold is documented.
Secrets access logs	DOC-19 >=90 days	Operational log archive meets minimum without secret values.
Media originals	DOC-17/19 tombstone/deletion safety	Historical backup versions remain private and inaccessible; backup expiry completes physical recovery-window removal.
Sessions/tokens	DOC-19 TTL/purge rules	Do not restore expired/revoked bearer authority from backup snapshots.
Canonical DB records	DOC-08 retention classes	PITR retention does not create an Admin restore feature or change logical hard/soft-delete semantics.

32.1 Media tombstone physical-purge horizon

Stage	Exact v1 rule
Public/tombstone effect	Immediate on accepted DOC-17 delete: media is DELETED, delivery revoked and CDN purge/invalidation begins.
Primary private blob purge	Eligible 7 days after tombstone only after authoritative usage recheck, no legal/incident hold, and current media row still DELETED.
Backup residual	A physically purged object may remain only in inaccessible encrypted backup/version recovery material until the 35-day backup window expires.
Restore safety	Finding a historical blob never reactivates media or makes it public; current DB status/site ownership and DOC-17 eligibility must still pass.

33. Load and Performance Test Specification

DOC-21 later owns the complete executable QA system. DOC-20 defines the performance scenarios and pass thresholds that DOC-21 must automate without weakening tenant/security assertions.

ID	Scenario	Pass gate
LT20-01	Public steady 300 RPS / 30m	All latency + availability SLOs; zero cross-site data leakage.
LT20-02	Public burst 600 RPS / 5m	No correctness loss; autoscale without DB connection ceiling breach.
LT20-03	Hot tenant 60% traffic	Other sites maintain >=90% of their baseline throughput and stay within 1.5x normal p95 latency.
LT20-04	Admin 25 RPS + mutations	No lost updates; expected_version conflicts remain deterministic.
LT20-05	140 simultaneous due publishes	>=99% within 60s; no early/duplicate publish.
LT20-06	10k worker backlog	Priority queues meet convergence SLOs; one tenant does not monopolize >50% when others

ID	Scenario	Pass gate
		wait.
LT20-07	Redis restart	No authorization/site leak; caches rebuild; duplicate jobs remain safe.
LT20-08	API replica termination under load	Availability stays within SLO; no sticky-session dependency.
LT20-09	Search outage/recovery	Non-search public pages stay available; search degrades/rebuilds same-site.
LT20-10	DB failover/connection reset	Ambiguous writes do not double-commit; idempotency/version checks protect replay.
LT20-11	Media transform backlog	Publish/cache queues unaffected; private original never public fallback.
LT20-12	PITR restore drill	RPO/RTO + RLS/isolation/audit integrity pass.

34. Capacity Planning and Scale Triggers

Resource	Planning rule
API CPU	Plan steady-state at <=50% average CPU under expected daily peak so one-replica failure/autoscale lag has headroom.
API memory	Steady-state <=65% pod/instance memory; critical at >85% sustained or any OOM restart.
DB connections	Keep configured aggregate pools <=70% max_connections and observed peak <=80% of that application budget.
DB storage	Warn at 70%, critical at 85%; include WAL/PITR growth and temporary index/migration headroom.
Redis memory	Steady <=70% usable; capacity increase before 75% sustained.
Broker	Maintain >=2x peak 15-minute accepted-message volume in safe disk/quota headroom.
Object storage	Capacity is elastic/provider-backed; monitor private originals separately from derived variants and backup replicas.
Search	Provision so single-site full reindex meets <=30m objective while serving ordinary queries.
Workers	Maintain at least 30% spare runnable concurrency at expected peak after respecting downstream connection limits.

35. Failure and Degradation Matrix

Dependency	Allowed degradation	Forbidden behavior
PostgreSQL	Required canonical operations return controlled 503/unavailable; replicas removed from readiness	Serve stale cross-site data, accept uncommitted writes, disable RLS
Redis	Cache misses/fallback authoritative reads; worker duplicate risk handled by fences	Treat cache as authorization, serve tenant-blind key, restore stale lease as truth
Search	Search page/service unavailable/degraded; other public content may continue	Fallback to all-site DB search or leak private/unpublished content
Broker	Async acceptance operations requiring durable enqueue fail before 202; committed ordinary mutations rely reconcile	Claim accepted when broker did not durably accept
Object store	Media-dependent operations degrade; existing cached safe variant may serve if still eligible	Expose private original or wrong-site object

Dependency	Allowed degradation	Forbidden behavior
CDN	Origin fallback only through deployment-safe canonical path	Ignore host/site identity or security headers
Secrets/security store	Protected/auth operations fail closed	Bypass MFA/session/CSRF/step-up
Observability backend	Service continues where safe; local buffering bounded	Block canonical write indefinitely or log sensitive body as fallback

36. Security, Privacy and Observability Boundaries

- Performance instrumentation must preserve DOC-19 minimization: no passwords, tokens, raw private URLs, ArticleBody or uploaded bytes in metrics/traces/logs.
- Backups are encrypted, access-controlled, audited operational assets. There is no new Admin backup/restore page or download endpoint.
- Load tests against production require sanitized/synthetic inputs and explicit operational approval; do not brute-force authentication or create persistent test content without cleanup authority.
- No performance cache may outlive an explicit security/session/site lifecycle revocation in a way that grants access. Security correctness has zero tolerance for stale allow decisions.
- Backup restore and DR procedures never create hidden Super Admin/MFA bypass. Owner recovery remains DOC-19.
- Observability site_id labels are internal operational identifiers only; public responses and cross-tenant dashboards remain governed by DOC-13/15/19 non-disclosure.

37. Cross-Document Consistency Audit

Authority	DOC-20 audit conclusion
DOC-01	PASS - shared scale without changing 14-site product model.
DOC-02	PASS - no permission/role change; capacity controls do not authorize.
DOC-03	PASS - existing features only; 100,000+ Post direction operationalized.
DOC-04	PASS - SiteContext/RLS/cache/search/worker isolation preserved under scale.
DOC-05/06	PASS - no new Public/Admin page for metrics/backup/health.
DOC-07	PASS - expected_version/publish/schedule semantics unchanged.
DOC-08	PASS - no table/column/index removed; pool budgets and PITR sit above frozen schema.
DOC-09	PASS - canonical host/routing remains authoritative through CDN/DR.
DOC-10/11	PASS - page flow/error states preserved; stale frontend results remain discarded.
DOC-12/13	PASS - no visual/PageData field changes; caches project canonical response shapes.
DOC-14	PASS - 14 frontends remain independently deployable; shared API scale has no site-to-site imports.
DOC-15 RC7	PASS - 112-operation HTTP contract/rate ceilings unchanged; health probes explicitly operational/non-business.
DOC-16 RC4	PASS - search/ranking/sitemap algorithms unchanged; only budgets/recovery time defined.
DOC-17 RC2	PASS - private original/native_webp/variant/CDN safety unchanged.
DOC-18 RC2	PASS - queue/job/retry/dedupe/Redis key semantics unchanged; DOC-20 only sizes and alerts.
DOC-19 RC2	PASS - security/session/audit/privacy retention preserved; DOC-20 supplies platform-wide SLO/DR thresholds.

AUDIT RESULT

No genuine blocking contradiction or missing upstream dependency was found for DOC-20 scope. Deliberate downstream implementation dependencies remain: concrete cloud/database/Redis/broker/search/observability products and deployment wiring belong DOC-23/24, while executable end-to-end load/isolation test automation belongs DOC-21.

38. Traceability Registry

Trace ID	Source	Locked DOC-20 requirement	Home
TR20-001	DOC-01/14	Horizontal API scale preserves one shared backend and 14 isolated frontend identities	Secs 3,6
TR20-002	DOC-04	SiteContext isolation and pooled-connection safety under load	Secs 7-10
TR20-003	DOC-08	Pool/index/query budgets without schema drift	Secs 8-10
TR20-004	DOC-15 RC7	Latency/availability around exact HTTP operations; no rate-ceiling change	Secs 4-6,24
TR20-005	DOC-17 RC2	CDN/media performance with private-original safety	Secs 12,17,27
TR20-006	DOC-18 RC2	Worker scaling around frozen queues/retries/Redis semantics	Secs 13-18,28
TR20-007	DOC-19 RC2	Security telemetry/retention/incident constraints	Secs 20-23,32,36
TR20-008	DOC-08/19	PITR/backup/retention operational authority	Secs 25-32
TR20-009	PDF-21	Performance scenarios become executable release tests	Sec 33
TR20-010	PDF-23/24	Concrete provider/runtime/deployment must satisfy targets	Sec 40

39. Comprehensive Acceptance Criteria

ID	Release assertion
QA20-001	All performance optimizations preserve trusted SiteContext, server authorization and RLS; no tenant-blind fallback exists.
QA20-002	API replicas are stateless for authoritative request/session/tenant/business state and any replica can serve a valid request.
QA20-003	Aggregate API+worker pool maxima are <=70% of PostgreSQL max_connections with >=30% reserved headroom.
QA20-004	API replica pool max is 20 and worker process pool max is 10 unless DOC-20 is explicitly revised.
QA20-005	Every site-owned transaction establishes SET LOCAL app.site_id and returns no persistent tenant state to the pool.
QA20-006	Public warm-cache p95 <=200ms and p99 <=500ms under LT20-01.
QA20-007	Public uncached p95 <=600ms and p99 <=1200ms under LT20-01.
QA20-008	Public search p95 <=800ms and p99 <=1500ms under reference load.
QA20-009	Admin read p95 <=700ms/p99 <=1500ms; non-media write p95 <=1000ms/p99 <=2000ms.
QA20-010	Auth login/MFA p95 <=1500ms/p99 <=3000ms without weakening Argon2id/MFA.
QA20-011	Async command acceptance p95 <=500ms/p99 <=1000ms when required

ID	Release assertion
	dependencies are healthy.
QA20-012	Media synchronous acceptance p95 <=10s/p99 <=20s in the reference image workload while still verifying native_webp.
QA20-013	Public service sustains 300 RPS for 30m and 600 RPS for 5m with zero cross-site leakage.
QA20-014	Hot-tenant test gives one site 60% traffic while other sites retain >=90% baseline throughput and <=1.5x baseline p95 latency.
QA20-015	Horizontal API scaling never causes configured DB pool totals to exceed the 70% connection budget.
QA20-016	Optional read replicas, if used, are never used for authorization, mutations, Preview, expected_version or lifecycle-critical reads.
QA20-017	Baseline DOC-08 indexes remain installed; no runtime/ad-hoc index mutation is used as a performance shortcut.
QA20-018	Interactive SQL timeouts are 5s read/10s write and timeout causes no partial commit.
QA20-019	Background SQL is chunked so individual statements stay <=30s target timeout.
QA20-020	Deadlock/serialization retries occur only through frozen idempotency/revalidation rules.
QA20-021	Redis steady memory remains <=70% target and unexpected eviction is alertable.
QA20-022	Redis outage never grants access or mixes tenant cache identity; authoritative fallback remains same-site.
QA20-023	Repeated eligible public-route synthetic workload reaches >=90% cache hit after warm-up.
QA20-024	CDN cache identity includes tenant host/site; same path on two sites never shares a wrong response.
QA20-025	CDN stale/error behavior never bypasses site disable/archive, media revocation or Preview/private eligibility.
QA20-026	When multiple sites have queued work, no one site consumes >50% shared queue concurrency for >60s unless serialization makes others unrunnable.
QA20-027	q18.publish oldest due age alert fires at >60s and due publication never occurs early to meet SLO.
QA20-028	140-Post schedule spike publishes >=99% eligible Posts within 60s and 100% within 5m absent dependency outage/intentional validation block.
QA20-029	Incremental search convergence p95 <=30s/p99 <=120s under healthy dependencies.
QA20-030	100k-document single-site full reindex completes <=30m and activation remains atomic.
QA20-031	All-site search rebuild after loss completes <=4h without cross-site index mixing.
QA20-032	Sitemap dirty-to-publish p95 <=5m/p99 <=15m and 25k shard/ordering rules stay intact.
QA20-033	Remaining media variants converge p95 <=5m/p99 <=15m; media backlog does not starve publish/cache queues.
QA20-034	Media CDN revoke/purge p95 <=30s/p99 <=2m without exposing private original.
QA20-035	Public cache invalidation p95 <=5s/p99 <=30s from authoritative change under healthy dependencies.
QA20-036	Health probes are operational only and do not create undocumented /api/v1 business endpoints.
QA20-037	API readiness requires critical DB/config/security prerequisites but does not become globally unready for unrelated derived-service failure.
QA20-038	Readiness uses 3 consecutive failures and 2 consecutive successes unless hard failure is reported.
QA20-039	Operational logs retain 30d; full traces 7d; raw-IP telemetry does not exceed

ID	Release assertion
	DOC-19 30d default; audit remains >=365d online; secrets access logs >=90d.
QA20-040	Metric labels never use post/media IDs, email, raw URL/query/IP/session/token or arbitrary exception text.
QA20-041	100% of 5xx/timeouts/restore/DLQ/destructive security-sensitive operations are trace-sampled without bodies.
QA20-042	Unexpected 5xx >5%/5m produces critical alert; >2%/5m warning.
QA20-043	DB pool >95%/5m or total connections >90% max produces critical alert and blocks further scale-out.
QA20-044	PITR/WAL archive lag >300s or any backup failure produces critical alert.
QA20-045	No successful base backup within 26h is critical.
QA20-046	audit_write_failure_total >0 is critical as required by DOC-19 signal semantics.
QA20-047	Production PostgreSQL PITR is enabled with base backup <=24h cadence and continuous WAL archiving.
QA20-048	PITR recoverable database window is 35 days and backups are encrypted/access controlled.
QA20-049	Quarterly PITR drill restores to a random point within 7d with measured RPO <=5m and RTO <=60m.
QA20-050	PITR validation proves RLS/FORCE RLS, tenant separation, permission seeds, Post states/versions and audit integrity before readiness.
QA20-051	Private media originals meet <=15m recovery-point target and <=2h recovery-time target.
QA20-052	Historical recovered originals remain private and cannot become PublicMedia fallback.
QA20-053	Redis/search/sitemap/media variants are treated as rebuildable; stale snapshots are never written back as canonical truth.
QA20-054	Catastrophic session-store loss does not resurrect stale sessions; affected users reauthenticate.
QA20-055	Broker durable acceptance survives normal node/process restart and catastrophic recovery target is <=5m RPO/<=30m RTO.
QA20-056	Restore sequence brings canonical DB/security ownership online before rebuilding tenant-derived caches/search/sitemaps/media variants.
QA20-057	DR restore never revives revoked secrets/session authority or bypasses DOC-19 owner recovery.
QA20-058	Quarterly media recovery drill uses multiple sites and same-filename collisions and still restores correct owner objects.
QA20-059	Quarterly empty-Redis/search/sitemap rebuild completes within the RTO matrix solely from authoritative sources.
QA20-060	Full platform DR game day occurs at least every 6 months and validates all 14 sites plus Central Admin/security/background convergence.
QA20-061	Backup copies that outlive active logical deletion remain inaccessible and expire within the 35-day backup recovery window absent legal hold.
QA20-062	Load test faults include API replica death, Redis restart, broker latency/restart, search outage, object-store transient failure and DB failover.
QA20-063	Load tests never weaken security/rate limits/RLS to obtain target numbers.
QA20-064	Performance telemetry outage cannot cause request bodies/secrets to be logged elsewhere or block canonical writes indefinitely.
QA20-065	Capacity planning keeps steady API CPU <=50%, memory <=65%, Redis <=70% and enough spare worker capacity to absorb one hot tenant.
QA20-066	DB/storage capacity alerts reserve growth/WAL/index/migration headroom and never wait for disk-full failure.
QA20-067	Cross-document audit has no unresolved blocker; any optimization that requires upstream contract change triggers controlled reopen.

ID	Release assertion
QA20-068	No new feature, permission, Page ID, API endpoint, DB table/column, queue/job type, media variant, search rule or security bypass is introduced by DOC-20.
QA20-069	Metrics retain 1-minute series 30d, 5-minute rollups 180d and 1-hour rollups 13 months without high-cardinality/privacy drift.
QA20-070	DELETED media private primary blobs become purge-eligible after 7d only after current usage/legal-hold/status recheck; backup residual expires within 35d.
QA20-071	DOC-21 may automate/expand these assertions but may not weaken numeric SLO/RPO/RTO gates.
QA20-072	DOC-23/24 may choose vendors/instance types/topology only if all DOC-20 performance/recovery/observability behavior remains satisfied.

40. Downstream Dependencies and Freeze-Readiness

Later authority	May define	May not change without reopening DOC-20
PDF-21 QA	Executable load/chaos/recovery harness, datasets, automation	Numeric targets, tenant-fairness and RPO/RTO expected outcomes
PDF-23 Engineering	Concrete DB/Redis/broker/search/observability/cloud products, SDKs and process topology	Statelessness, pool budgets, health semantics, backup classes or correctness boundaries
PDF-24 Deployment	Environment sizing, autoscaler configuration, probes, rollout, migration, backup jobs and restore commands	SLOs/RPO/RTO, PITR adoption, 35d recovery window, staged restore verification
PDF-22 Traceability	Full requirement-to-test mapping	Authority ownership or acceptance meaning

40.1 Freeze-readiness checklist

- FZ20-01 Performance authority boundary accepted.
- FZ20-02 Reference 14-site/100k-Post load profile accepted.
- FZ20-03 Public/Admin/Auth/API latency targets accepted.
- FZ20-04 Stateless API/horizontal scaling rules accepted.
- FZ20-05 Noisy-neighbor fairness rule accepted.
- FZ20-06 PostgreSQL aggregate 70% connection budget and pool caps accepted.
- FZ20-07 DB statement/query/lock thresholds accepted.
- FZ20-08 Redis memory/latency/eviction policy accepted.
- FZ20-09 CDN/edge/origin protection rules accepted.
- FZ20-10 Worker queue scaling and derived convergence targets accepted.
- FZ20-11 Health readiness/liveness semantics accepted.
- FZ20-12 Logs/metrics/traces retention and sampling accepted.
- FZ20-13 Alert registry/thresholds accepted.
- FZ20-14 Monthly availability SLO/error-budget policy accepted.
- FZ20-15 PostgreSQL PITR adoption + 24h base backup + 35d window accepted.
- FZ20-16 RPO/RTO matrix accepted.
- FZ20-17 Object original recovery and derived-state rebuild policy accepted.
- FZ20-18 Quarterly restore drills and six-month full DR game day accepted.
- FZ20-19 72 QA20 assertions accepted as release-blocking baseline.
- FZ20-20 Cross-document audit has zero unresolved blockers and downstream vendor/deployment choices remain constrained.

REVIEW STATUS

DOC-20 v1.0-RC1 is a complete review candidate. No genuine blocking contradiction or missing upstream dependency was found in DOC-01 through DOC-19 for the responsibilities owned here. Formal freeze still requires explicit project approval.

Appendix A. Performance Target Quick Reference

Area	Target
Public warm cache	p95 200ms / p99 500ms
Public uncached	p95 600ms / p99 1.2s
Public search	p95 800ms / p99 1.5s
Admin read	p95 700ms / p99 1.5s
Admin write	p95 1s / p99 2s
Auth	p95 1.5s / p99 3s
Async accept	p95 500ms / p99 1s
Media sync phase	p95 10s / p99 20s
Public steady/burst	300 RPS 30m / 600 RPS 5m
Availability	99.90% public/admin/auth; public search 99.50%
Cache invalidation	p95 5s / p99 30s
Search convergence	p95 30s / p99 120s
Sitemap	p95 5m / p99 15m
Media variants	p95 5m / p99 15m
Scheduled publish	p95 +10s / p99 +60s

Appendix B. Alert Threshold Registry

ID	Severity	Threshold
ALT20-01	Critical	Unexpected 5xx >5% for 5m on public/admin/auth
ALT20-02	Warning	Unexpected 5xx >2% for 5m
ALT20-03	Warning	Applicable p95 latency exceeds SLO for 10m
ALT20-04	Critical	Applicable p99 latency exceeds 2x SLO for 5m
ALT20-05	Warning	DB pool >80% for 10m or pool-wait p95 >100ms
ALT20-06	Critical	DB pool >95% for 5m or total DB connections >90% max
ALT20-07	Critical	PITR/WAL archive lag >300s or backup job fails
ALT20-08	Critical	No successful base backup in 26h
ALT20-09	Warning	Redis memory >75% for 10m
ALT20-10	Critical	Redis memory >90% or unexpected eviction observed
ALT20-11	Critical	q18.publish oldest eligible due job >60s
ALT20-12	Warning	q18.search or q18.media oldest job >5m
ALT20-13	Critical	q18.search or q18.media oldest job >15m
ALT20-14	Critical	Any new publish DLQ entry or cache invalidation DLQ entry
ALT20-15	Warning	Any other DLQ depth >0 for 10m
ALT20-16	Critical	audit_write_failure_total increases above 0
ALT20-17	Critical	Restore drill fails checksum/integrity/isolation

ID	Severity	Threshold
		verification
ALT20-18	Critical	Cross-site denial >=5 from same principal/source in 5m or >=50 platform-wide in 5m
ALT20-19	Warning	CDN origin error >1% for 5m
ALT20-20	Critical	Health-ready capacity <50% of required replicas for 5m

Appendix C. RPO/RTO and Backup Registry

ID	Asset	RPO	RTO
DR20-01	PostgreSQL canonical data	<=5m	<=60m
DR20-02	Private media originals	<=15m	<=2h
DR20-03	API/Admin/Auth processes	0 canonical data loss - stateless	<=30m after infrastructure available
DR20-04	Broker accepted work	<=5m catastrophic / 0 for normal HA restart	<=30m
DR20-05	Redis cache/coordination	No canonical RPO	<=15m service restore; lazy rebuild continues
DR20-06	Public search index	Rebuildable	<=2h site service; <=4h all-site rebuild
DR20-07	Sitemap artifacts	Rebuildable	<=2h all ACTIVE sites
DR20-08	Media derived variants	Rebuildable	<=4h backlog recovery after originals available
DR20-09	Admin/session ephemeral state	Intentionally not restored stale	<=60m auth service; users may re-login

Backup control	Frozen value
PostgreSQL PITR	ADOPTED
Base backup	<=24h
WAL archive lag	<=5m
DB recovery window	35d
DB RPO/RTO	<=5m / <=60m
Private original RPO/RTO	<=15m / <=2h
PITR drill	Quarterly
Full platform DR game day	Every 6 months

Appendix D. One-Page Developer Execution Reference

Layer	Execution lock
API scale	Stateless replicas; preserve SiteContext/authorization; autoscale only inside DB/Redis/dependency budgets.
Database	Primary canonical; pools <=70% max_connections aggregate; SET LOCAL tenant every site transaction; no stale tenant connection.
Redis	DOC-18 keys/TTLs unchanged; derived only; steady memory <=70%; loss may duplicate work, never grant access.
Workers	Scale per queue age/depth; retain DOC-18 retries/dedupe/leases; hot site <=50% shared concurrency when others wait.
CDN/media	Host/site cache identity; Admin/Preview no-store; private original never public; purge/revoke meets DOC-17.
Observability	Safe logs 30d; traces 7d; metrics low-cardinality; exact critical thresholds in Appendix B.
Backup	PostgreSQL PITR adopted: base <=24h + continuous WAL + 35d window. Private originals protected. Derived state rebuilt.
DR	DB RPO <=5m/RTO <=60m; private originals <=15m/<=2h; quarterly restore drills; full game day every 6 months.
Release gate	Pass reference 14-site/100k-Post load, failure and restore scenarios with zero isolation/security/correctness regression.
Availability	99.90% monthly Public/Admin/Auth; public Search 99.50%; intentional auth/rate/redirect/not-found outcomes excluded from outage accounting.
Derived convergence	Cache p99 30s; search p99 120s; sitemap/media p99 15m; publish p99 60s under healthy dependencies.
Telemetry retention	Operational logs 30d; traces 7d; metrics 1m/30d, 5m/180d, 1h/13mo; audit/security retention remains DOC-19 bounded.
Media deletion	Logical revoke immediately; private primary blob purge eligible after 7d recheck; backup residual expires by 35d absent hold.